

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 — 2024

Data: Global Carbon Budget 2025 • NASA / NOAA Mauna Loa Observatory

Prepared for Policy Analysts & Climate Negotiators

Atmospheric CO₂ Reaches 429 ppm

50% Above Pre-Industrial Levels

429

parts per million

NOAA Mauna Loa — February 2026

~280 ppm

Pre-industrial baseline

1.5×

Faster than last ice-age r

~200 years

Time span of current
increase

Ground-based and satellite measurements confirm the sharp acceleration, linked primarily to the burning of coal, oil, and natural gas.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold

6 Billion Tonnes in 1950 → 35+ Billion Tonnes Today

6 Bt

1950

Post-war industrial take-off

20 Bt

1990

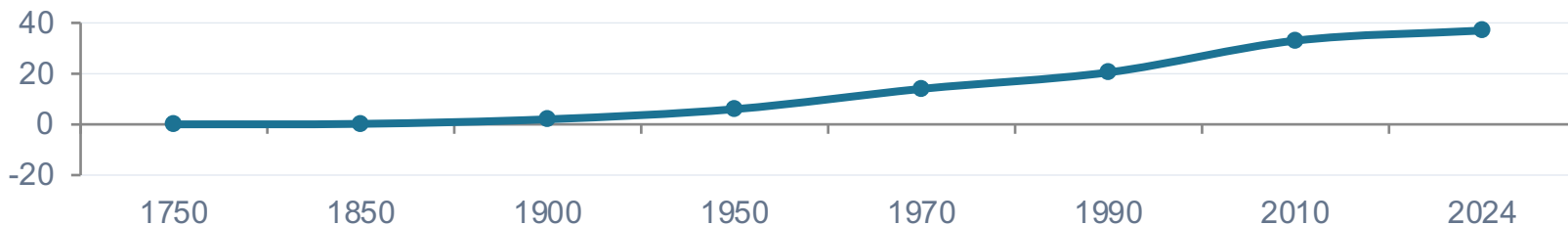
Nearly quadrupled in 40 years

35+ Bt

2024

Growth slowed, but not yet peaked

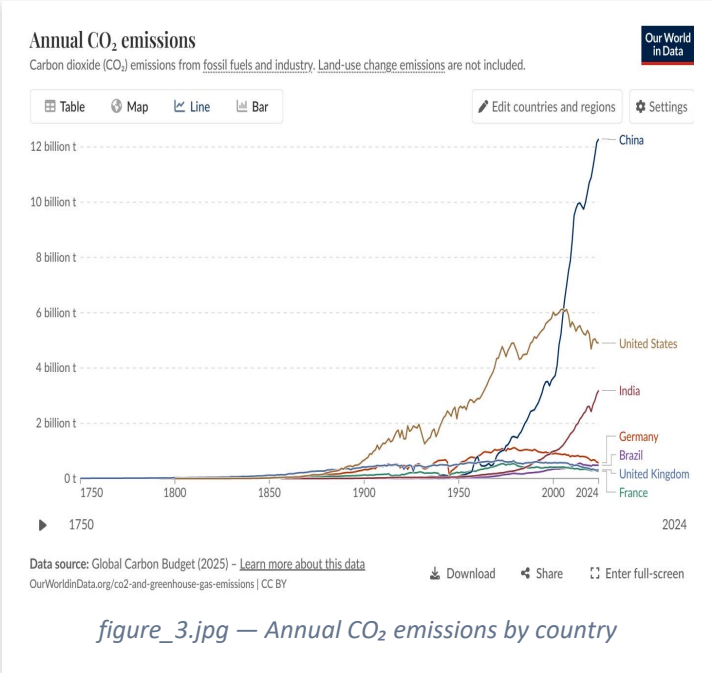
Reference: figure_2.jpg — Annual CO₂ emissions trajectory (1750–2024)



China Now Emits Over 12 Billion Tonnes

More Than Double the United States

Country / Region	Annual CO ₂ (Bt)	Global Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
The US peaked near 6 Bt around 2005 and has since declined to roughly 5 Bt. India's rise is rapid.			
France	~0.3	~0.8%	Declining



Asia Now Accounts for ~50% of Global Emissions

US + Europe Combined Have Fallen Below One-Third

1900

Europe + United States

> 90%

of global CO₂ emissions

2024

Asia

~ 50%

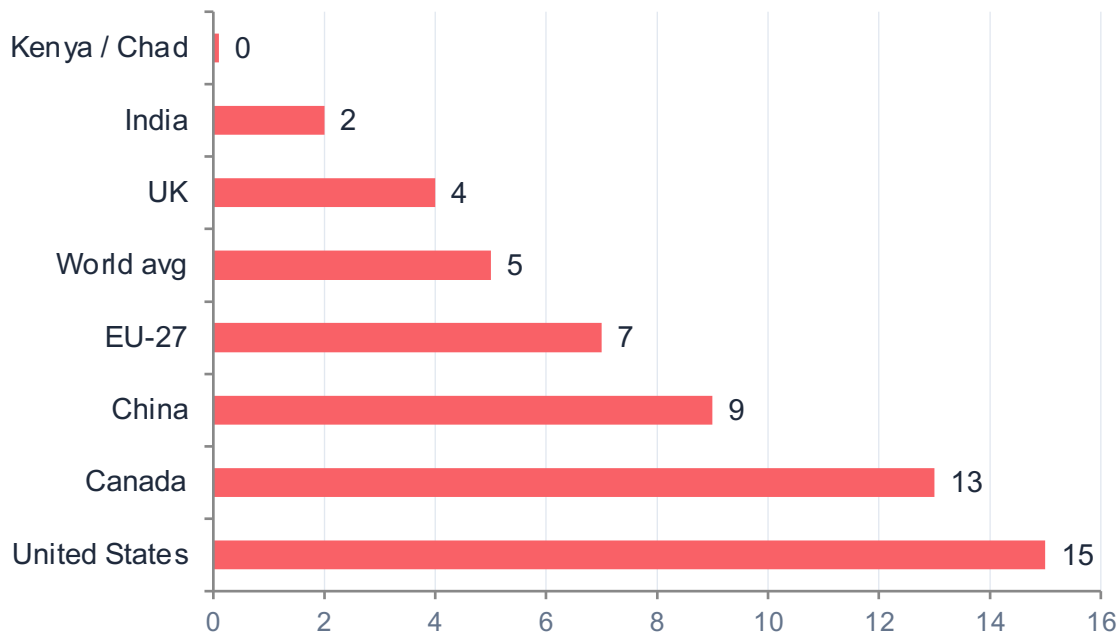
of global CO₂ emissions

Africa and South America each contribute only 3–4%, comparable to international aviation and shipping combined.

US + Europe today: < 33% • Asia: ~50% • Rest of world: ~17–20%

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad

A 150x Disparity



150x

gap between highest and
lowest per capita emitters

*The average American emits in
under two days what the
average person in Chad or
Niger emits in an entire year.*

Reference: figure_1.jpg — Per capita CO₂ emissions over time

2024 Snapshot: US and Europe Cut Emissions

Russia and Southeast Asia Increase

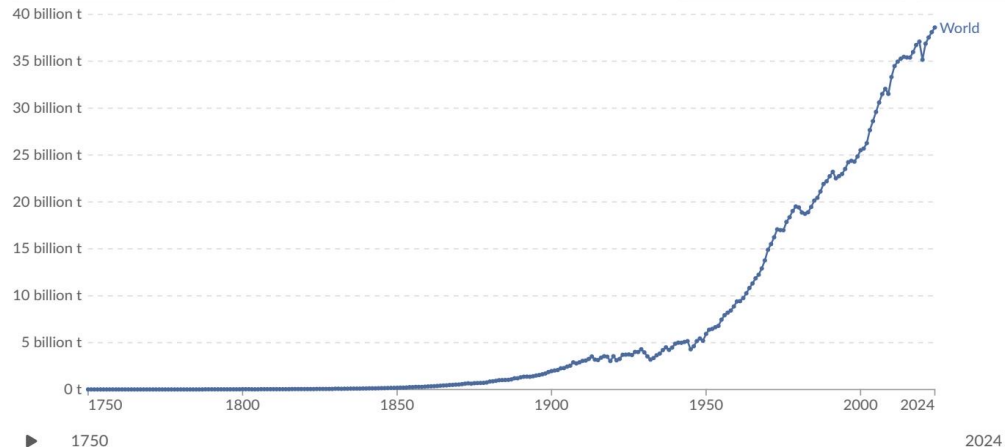
Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data

Table Map Line Bar

Edit countries and regions Settings



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

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Regional Patterns



United States & much of Europe
-1% to -5%



Russia, parts of SE Asia, Africa
+2% to +10%



Some South American nations
> +10%

Mixed picture: declines in mature economies, increases in emerging markets.

figure_4.jpg — Annual % change in CO₂ emissions, 2024

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Policy choices — not just prosperity — determine emissions outcomes

	France	United Kingdom	Germany	Netherlands
Nuclear share	High	Moderate → growing	Low	Low
Renewable share	High	High	Moderate	Moderate
Fossil fuel share	Low	Low	High	High
Per capita CO ₂	~4 t	~4 t	Higher	Higher

Insight: Countries with similar living standards can differ significantly in per capita emissions because of their energy mix — higher nuclear and renewable shares translate directly to lower CO₂.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
- China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.
- The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.
- Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
- Energy mix and policy choices matter: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.